Dirichlet's theorem

Sophia MAI

April 2025



Abstract

In this discussion, we prove the formula for the analytic density of primes congruent to $a \mod m$ and deduce Dirichlet's theorem. We then apply this result to demonstrate the existence of solutions to equation $X^2 = a$. The notions of the Riemann ζ-function, the infinitude of primes congruent to 1 or 3 modulo 4, characters of finite abelian groups, Dirichlet characters and Dirichlet L-functions have already been introduced and established by my classmates in Andrea Conti's course.

The distribution of prime numbers is a central topic in number theory. While primes seem scattered, deeper analysis reveals patterns influenced by arithmetic structure. In this study, we examine how primes are distributed with respect to congruence conditions, using tools like Dirichlet characters and L-functions. We conclude with an application to quadratic residues, showing how these concepts lead to concrete number-theoretic results.

Dirichlet's theorem

The following definitions follow [Ked07].

Definition 1.1. An arithmetic progression is *eligible* if it has the form $m, m+N, m+2N, ...$ where $gcd(m, N) = 1$; it is equivalent to asking that two consecutive integers contain infinitely many primes.

Dirichlet theorem

Any eligible arithmetic progression of positive integers contains infinitely many primes.

In other words, for any integers $m \geq 2$ and $a \geq 1$ such that $gcd(a, m) = 1$, there are infinitely many prime numbers p satisfying $p \equiv a \mod m$.

To understand the structure of prime distribution, we first introduce the notion of density.

Natural and analytic density

Definition 2.1. For $S \subseteq T$ two sets of positive integers, with T infinite, the *upper natural density* and *lower natural density* of S in T are defined as

$$\limsup_{N\to\infty} \frac{\#\{n \in S : n \leq N\}}{\#\{n \in T : n \leq N\}},$$

$$\liminf_{N\to\infty} \frac{\#\{n \in S : n \leq N\}}{\#\{n \in T : n \leq N\}}.$$

If they coincide, we call the common value the *natural density* (or *asymptotic density*) of S in T.

Many interesting sets fail to have a natural density. We get a less restrictive notion of density by using Dirichlet series.

Definition 2.2. For $S \subseteq T$ two sets of positive integers, with $\sum_{n\in T} n^{-1}$ divergent, the *upper Dirichlet density* and *lower Dirichlet density* of S in T are defined as

$$\limsup_{s\to 1^+} \frac{\sum_{n\in S} n^{-s}}{\sum_{n\in T} n^{-s}},$$

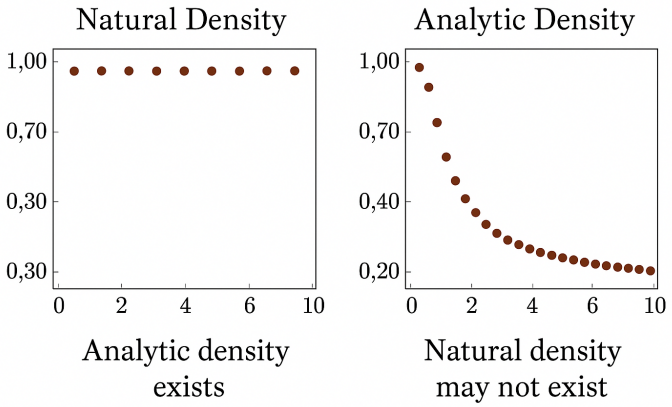
$$\liminf_{s\to 1^+} \frac{\sum_{n\in S} n^{-s}}{\sum_{n\in T} n^{-s}}.$$

If they coincide, we call the common value the *Dirichlet density* of S in T.

Remark 2.3. Any finite set has density 0 in any infinite set.

Remark 2.4. If A has a natural density k, the analytic density of A exists and is equal to k. On the other hand, there exist sets that have an analytic density but no natural density. It is the case, for example, of the set P^1 of prime numbers whose first digit (in the decimal system, say) is equal to 1. One sees easily, using the prime number theorem, that P^1 does not have a natural density and, on the other hand, Bombieri has shown a proof that the analytic density of P^1 exists (it is equal to $log_{10}2 = 0,301029995...$).
However, this "pathology" does not occur for the sets of prime numbers considered above: *the set of* $p \in P$ *such that* $p \equiv a \mod m$*, if* a *is prime to* m has a natural density equal to the analytic density.

This is why we will focus on analytic density.



Proposition 2.5. *Let P be the set of prime numbers. When s tends to 1 (s being real > 1 to fix the ideas) one has*

$$\sum_{p\in P} \frac{1}{p^S} \sim \log(\frac{1}{s-1}).$$

Proof: [Ked07] Recall that $\zeta(s)$ has a simple pole of residue 1 at $s = 1$, so as $s \to 1^+$, with $s \in \mathbb{C}$,

$$\zeta(s) \sim \frac{1}{s-1}$$

and then :

$$(s-1)\sum_{n=1}^{\infty} n^{-s} = 1 + o(1).$$

Hence if $T = \mathbb{N}$, then the Dirichlet density of S is given by

$$\lim_{s\to 1^+} (s-1)\sum_{n\in S} n^{-s}$$

if the limit exists.
Taking logarithms, we see that as $s \to 1^+$,

$$\log(\zeta(s)) + \log(s-1) = \log(s-1) + \sum_p \sum_{n=1}^{\infty} p^{-ns} = O(1).$$

We use for $\log(\zeta(s))$ the expression of $\zeta(s)$ in its product form.
We therefore have $\sum_p \sum_{n=2}^{\infty} p^{-ns} = O(1)$, so

$$\sum_p p^{-s} = -\log(s-1) + O(1).$$ ■

We use this equivalence to redefine the analytic density.

Remark 2.6. Let A be a subset of P. One says that A has for *density* a real number k when the ratio

$$(\sum_{p\in A} \frac{1}{p^s})/(\log \frac{1}{s-1})$$

tends to k when $s \to 1$.

We will focus on this form of density. Once we understand the analytic density, the next logical step is to explore how it leads to the density theorem for primes in arithmetic progressions.

Density theorem

Density theorem

Let $m \geq 1$ and let a be such that $pcd(a, m) = 1$. Let P_a be the set of prime numbers such that $p \equiv a \mod m$. The set P_a has density $1/\phi(m)$.

In other words, the prime numbers are "distributed evenly" among the various classes modulo m that are coprime to m.

Remark 3.1. The set P_a must be infinite, since it has a nonzero density — and, as stated in Remark 2.3, any finite set has density zero. Consequently, Dirichlet's theorem follows directly from the density theorem, confirming the infinitude of primes in eligible arithmetic progressions.

To prove the theorem, we have to study the behavior of the function

$$g_a(s) = \sum_{p \in P_a} \frac{1}{p^s},$$

with P_a the set of primes $p \equiv a \mod m$.

Remark 3.2. If P_a is finite, $g_a(s)$ is a finite sum, and so well-defined for all $s \in \mathbb{C}$. But $g_a(s)$ is not defined for $s = 1$. It follows that P_a is infinite.

Now, let's present some lemmas that will help us prove the density theorem. These are taken from [Ser77].

Let χ be a character of $G(m) = (\mathbb{Z}/m\mathbb{Z})^\times$. Put

$$f_\chi(s) = \sum_{p \neq m} \frac{\chi(p)}{p^s},$$

the series being convergent for $s > 1$.

Lemma 3.3. *If $\chi = 1$, then $f_\chi \sim \log(\frac{1}{s-1})$ for $s \to 1$.*

Indeed, f_s differs from the series $\sum \frac{1}{p^s}$ by a finite number of terms only.

Lemma 3.4. *If $\chi \neq 1$, f_χ remains bounded when $s \to 1$.*

Proof: We will use the product representation of the L-function in the computation of its logarithm (more precisely, the principal determination of the logarithm). For $Re(s) > 1$ each factor is of the form $\frac{1}{1-\alpha}$ with $|\alpha| < 1$. We define $\log(\frac{1}{1-\alpha})$ as $\sum_{n=1} \alpha^n n$.

$$L(s,\chi) = \prod \frac{1}{(1-\chi(p)p^{-s}}$$

$$\log(L(s,\chi)) = \sum \log(\frac{1}{1-\chi(p)p^{-s}}) = \sum_{n,p} \frac{\chi(p)^n}{np^{ns}}$$

with $Re(s) > 1$. We now split $\log L(s,\chi)$ into two parts:

$$\log L(s,\chi) = f_\chi(s) + F_\chi(s)$$

with

$$F_\chi = \sum_{p,n\geq 2} \frac{\chi(p)^n}{np^{ns}}.$$

$F_\chi(s)$ remains bounded near $s = 1$ because it converges absolutely (exponential decay). $\log L(s,\chi)$ is bounded near $s = 1$ because $L(s,\chi)$ is analytic and non-zero when near 1 when χ is non-principal. The potential singularity at $s = 1$ comes only from the principal character. Hence, the same holds for $f_\chi(s)$, which proves the lemma. ■

Let's return to the function $g_a(s)$ defined earlier and rewrite it using the expression for $f_\chi(s)$. We now know that $f_\chi(s)$ is bounded for $\chi \neq 1$ and equals $\log \frac{1}{s-1}$ for $\chi = 1$.

Lemma 3.5. *One has*

$$g_a(s) = \frac{1}{\phi(m)} \sum_\chi \chi(a)^{-1} f_\chi(s),$$

the sum being extended over all characters χ of $G(m)$.

Proof: The function $\sum \chi(a)^{-1} f_\chi(s)$ can be written, by replacing f_χ by its definition:

$$\sum_{p\neq m} (\sum_\chi \chi(a^{-1})\chi(p))/p^s.$$

But $\chi(a^{-1})\chi(p) = \chi(a^{-1}p)$. Using the orthogonality relations (where $\hat{G}$ is the dual of G and $n = Card(\hat{G})$)

$$\sum_{\chi\in\hat{G}} \chi(x) = \begin{cases} n & \text{if } \chi = 1 \\ 0 & \text{if } \chi \neq 1 \end{cases},$$

we have:

$$\sum_\chi \chi(a^{-1}p) = \begin{cases} \varphi(m) & \text{if } a^{-1}p \equiv 1 \pmod m \\ 0 & \text{otherwise} \end{cases}$$

Hence we find the function $\phi(m)g_a(s)$.

The density theorem is now evident : we have seen that $f_\chi(s) \sim \log \frac{1}{s-1}$ when $\chi = 1$, and that $f_\chi(s)$ bounded for $\chi \neq 1$. It follows that

$$g_a \sim \frac{1}{\phi(m)} \log \frac{1}{s-1},$$

which implies that the density of the set P_a is $\frac{1}{\phi(m)}$. ∎

Having established the density theorem, we now turn to an important application concerning quadratic residues

Application

This section is based on applications found in [Ser77].

Proposition 4.1. *Let a be an integer which is not a square. The set of prime numbers p such that $\left(\frac{a}{p}\right) = 1$ has density $\frac{1}{2}$.*

This concerns the frequency with which a given integer a is a quadratic residue modulo p when considering the set of all prime numbers.

To recall the Legendre symbol, $\left(\frac{a}{p}\right)$ is defined as follows:

$$\left(\frac{a}{p}\right) = \begin{cases} 0 & \text{if } p \mid a \\ 1 & \text{if } a \text{ is a quadratic residue modulo } p \\ -1 & \text{if } a \text{ is a non-quadratic residue modulo } p \end{cases}$$

In this context, a is fixed, and the set of primes p that divide a is finite. Therefore, the density of primes p dividing a is 0. Implicitly, we are restricting our attention to primes p such that $p \nmid a$, i.e., where the Legendre symbol $\left(\frac{a}{p}\right)$ takes values ± 1.

I would like to recall a proposition that was presented in the previous talk by my classmate:

Proposition 4.2. *Let a be a non-zero square-free integer and let $m = 4|a|$. Then there exists a unique character χ_a modulo m such that $\chi_a = \left(\frac{a}{p}\right)$ for all prime numbers p not dividing m. One has $\chi_a^2 = 1$ and $\chi_a \neq 1$ if $a \neq 1$.*

Proof: (for Proposition 4.1.) We can assume that a is square-free. Let $m = 4|a|$, let χ_a be the character modulo m as defined in the previous proposition and let $H \subset G(m)$ be the

kernel of χ_a in $G(m)$. If p is a prime number not divisible by m, let $\bar{p}$ be its image in $G(m)$. We have $\left(\frac{a}{p}\right) = 1$ if and only if $\bar{p}$ is containded in H. By the theorem of density, the set of prime numbers verifying this condition has for density the inverse of the index $[G(m) : H]$, that is to say $\frac{1}{2}$. ∎

Since χ_a is a non-trivial Dirichlet character of order 2 (i.e. $\chi_a(p) = \pm 1$ for $p \nmid m$, and $\chi_a^2(p) = 1$), its kernel H has index 2 in $G(m)$. Therefore, the density is $\frac{1}{2}$, as claimed.

Corollary 4.3. *Let a be an integer. If the equation $X^2 - a = 0$ has a solution modulo p for almost all $p \in P$, it has a solution in $\mathbb{Z}$.*

This is an example of local-global principle. If an integer behaves like a square modulo almost all primes, it must be a square in $\mathbb{Z}$. If a is not as square, then $\left(\frac{a}{p}\right) = 1$ for only half of the primes p. Therefore, if $\left(\frac{a}{p}\right) = 1$ for almost all primes p, then a must be a square in $\mathbb{Z}$.

Remark 4.4. There are analogous results for other types of equations. For instance:

i) Let $f(x) = a_0X^n + ... + a_n$ be a polynomial of degree n with integer coefficients, which is irreductible over $\mathbb{Q}$. Let K be the field generated by the roots of f (in an algebraically closed extension of $\mathbb{Q}$) and let $N = [K : \mathbb{Q}]$. One has $N \geq n$. Let P_f be the set of prime numbers p such that f "decomposes completely modulo p", i.e. such that all the roots of f modulo p belong to F_p. One can prove that P_f has density $\frac{1}{N}$.(The method is analogous to that of Dirichlet theorem - one uses the fact that the zeta function of the field K has simple pole at $s = 1$.) One can also give the density of the set P'_f of p such that the reduciton of f modulo p has at least one root in F_p; it is a number of the form $\frac{a}{N}$ with $1 \leq q < N$ (setting aside the trivial case where $n = 1$).

ii) More generally, let $\{f_\alpha(x_1, ..., x_n)\}$ be a family of polynomials with integer coefficients and let $\mathbb{Q}$ be the set of $p \in P$ such that the reductions of f_α modulo p have a common zero in $(F_p)^n$. It can be proved that $\mathbb{Q}$ has a density; moreover this density is a rational number and is zero only if $\mathbb{Q}$ is finite.

In the first example, an irreducible polynomial over $\mathbb{Q}$ rarely splits completely modulo a prime—this happens for only $\frac{1}{N}$ of the primes. Still, it may have roots modulo more primes, with the proportion determined by the associated number field. In the second case, if a system has no solutions modulo almost all primes, it likely has no global solution; but if it has solutions for infinitely many primes, this is meaningful and measurable via density.

We showed that primes congruent to $a \mod m$ with $\gcd(a, m) = 1$, have analytic density $\frac{1}{\phi(m)}$, confirming Dirichlet's theorem and the uniform distribution of primes among coprime residue classes. This highlights the key role of analytic tools like Dirichlet L-functions and character sums, which also help solve classical problems such as identifying when an integer is a quadratic residue for infinitely many primes.

References

[Ked07] K. Kedlaya. *Primes in arithmetic progressions*. Analytic Number Theory. Cambridge: MIT, 2007.

[Ser77] J.-P. Serre. *Cours d'Arithmétique, Deuxième Édition Revue et Corrigée*. Le Mathématicien. Paris: Presses Universitaires de France, 1977.